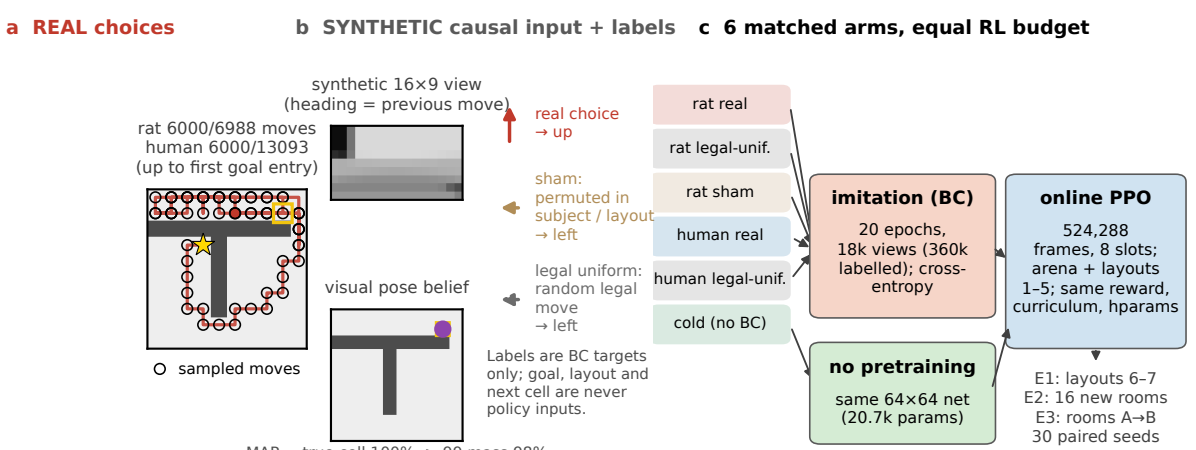

a REAL choices
b SYNTHETIC causal input + labels
c 6 matched arms, equal RL budget
synthetic 16×9 view
(heading = previous move)
rat 6000/6988 moves
human 6000/13093
(up to first goal entry)
real choice
→ up
sham:
permuted in
subject / layout
→ left
legal uniform:
random legal
move
→ left
visual pose belief
sampled moves
Labels are BC targets
only; goal, layout and
next cell are never
policy inputs.
MAP = true cell 100%; >.99 mass 98%
rat real
rat legal-unif.
rat sham
human real
human legal-unif.
cold (no BC)
imitation (BC)
20 epochs,
18k views (360k
labelled); cross-
entropy
no pretraining
same 64×64 net
(20.7k params)
online PPO
524,288
frames, 8 slots;
arena + layouts
1–5; same reward,
curriculum, hparams
E1: layouts 6–7
E2: 16 new rooms
E3: rooms A→B
30 paired seeds